NATHAN MANLEY

Cork, Ireland | Second Year Electrical & Electronic Engineering Student at UCC
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More information on my projects and experience available on my website.

EDUCATION

University College Cork — *2nd Year E&E Engineering — First Class Honours (1.1) | GPA: 3.75/4* 2025 - 2029

Quercus Talented Student Programme Innovation & Entrepreneurship Scholar 2026 - Present

Fully funded scholarship, awarded as recognition for entrepreneurial and innovative endeavours both on and off campus.

Glanmire Community College — *Leaving Certificate — 589/625 Points* 2025
Subjects: Applied Maths, Physics, Technology, Maths, English, History, Irish, French

WORK EXPERIENCE

Logitech — *Software UX Designer* 2026 - Present
- Creating and designing plugins for Logitech MX Keypad.

Logitech — *Software UX Design Intern* Summer 2026
- UX Design Explorations for upcoming PWS POVs.

ATMac — *Repair Technician* Summer 2023/2024
- Diagnosed and repaired various Apple laptops and desktops.
- Enhanced technical troubleshooting and component level repair skills.

TECHNICAL / ENGINEERING EXPERIENCE

Matheson LLP Hackathon 2026
- Winner of Matheson LLP hackathon, with 100s of attendees.
- Developed the Career Engine, a platform for CEOs to help recruit and retain employees through degree agnostic hiring and training practices.

WindEurope HackTheClimate Hackathon 2026
- Developed EZBreezy, a decision-support system that integrates with SCADA and Eirgrid API to advise TSO's on how to keep as much renewable energy on the grid as possible.
- Spent 3 days hacking in Microsoft Dublin developing UX in React and FastAPI.

National SDG Hackathon 2026
- Winner of National SDG Hackathon, UCC and Best Pitch Award, winning €4K in prize money.
- Developed Remento, an app for early-stage dementia tracking/detection for families of the elderly.

HackEurope 2026
- Developed WindSite, utilising Claude, Python, and React Web Development.
- WindSite is an intelligent platform that integrates a modular node system flow chart to make offshore wind turbine farm planning easier and smarter.

Patch 2024
- Selected as one of 30 innovators across Europe to complete the program in Dublin.
- Co-founded Gymificient, a platform providing actionable analytics to gym owners for machine acquisition and maintenance.
- Developed backend logic and analytics modules using OpenCV, in C++ and Python.

UCC Rocketry and Space Exploration Society — *Payload and Hardware Avionics Team Member* 2025–2026
- Programming camera telemetry and payload systems for MACH '26. AV Bay 3D Designer for EuRoC '26.

CanSat — *Lead Programmer, Lead 3D Can Design* 2023-2024
- Designed core satellite structure using Autodesk Fusion 360.
- Programmed main avionics systems in C++ (Arduino IDE).
- Optimized housing for electronics and produced parts on my personal 3D printer.

INDEPENDENT PROJECTS / ACHIEVEMENTS

- Telemetry Parsers: Captures and interprets UDP data from simulators like Gran Turismo 7 for integration with microcontrollers and dashboards. Various modules were programmed in C++, Python and Javascript.
- School Awards: Environmental and Innovation "Diograis" awards while attending GCC.

VOLUNTEERING

CoderDojo Glanmire
Joined as a programmer in 2014, receiving free mentorship. I became a mentor in 2023 to volunteer my time to teach today's young learners C++, Python, and HTML.

Gaisce 2023
Bronze Gaisce award.

INTERESTS

- Technical: Microcontrollers (ESP32/ESP8266), Embedded Systems, 3D Printing (Ender 3 V2).
- Personal: Weightlifting, Formula 1, Drifting.
- Academic: Electrical, Electronic & Mechanical Engineering. Rocketry and CanSat.

References available upon request